Colt M1911 / M1911A1 & Series 70/80

Colt's Manufacturing / US Ordnance • Government, Commander & Officer Models (.45 ACP, 9mm, 10mm Auto)

Manufacturer:	Colt's Manufacturing / US Ordnance	**Category:**	Pistols
Action Type:	Single-Action Semi-Automatic (Hammer-Fired)	**Era / Service:**	Military & Custom Benchmark (1911–Present)
Calibers:	.45 ACP, 9x19mm Parabellum, 10mm Auto, .38 Super	**Covered Models:**	Government Model (5"), Commander (4.25"), Officer's ACP (3.5"), M1911A1 USGI, Series 70 Gold Cup, Series 80

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Extractor Tension Tuning (25–28 oz) • Barrel Bushing Hand-Fitting • Sear & Hammer Hook Stoning (0.020") • Grip Safety Armorer Timing • Plunger Tube Staking

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
EXTRACTOR TENSION	25 to 28 ounces pull force to slide a loaded cartridge case down breech face
HAMMER HOOKS	Exact 0.020" depth, cut square at 90° with sharp corner (no radius on primary ledge)
SEAR PRIMARY ANGLE	Ground flat with escape angle / breakaway secondary cut (approx 30° relief on rear 1/3 of face)
BARREL BUSHING CLEARANCE	0.001" to 0.0015" fit to barrel OD; finger-tight bushing requiring wrench for match lockup
BARREL LINK TIMING	#3 link standard (0.278" center-to-center); lower lugs must rest evenly on slide stop cross pin
TRIGGER PULL TARGET	3.5 to 4.0 lbs (carry/duty) / 2.5 to 3.0 lbs (match bulls-eye target)
PLUNGER TUBE STAKING	Double internal flared rivet staking using dedicated 1911 staking pliers

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Failure to Extract / Double Feed (Type 2 Malfunction) **Spent casing left in chamber or pulled halfway out while new round tries to chamber beneath it**	• Loss of extractor claw spring tension (deflection under 20 oz). • Chipped or broken extractor claw tip. • Clocking extractor (flat firing pin stop allowing extractor to rotate in channel). • Severely fouled extractor tunnel packed with bullet lube and powder residue.	**Immediate Action: Lock slide back, strip magazine, cycle slide twice to clear chamber, reload. Gunsmith Action: Remove firing pin stop and slide extractor out. Clean channel. Inspect claw with 10x loupe. Bend extractor mid-body to increase spring deflection until a dummy round is held firmly against breech face with 25–28 oz pull.**
Stoppage 2: Hammer Follow / Accidental Slip to Half-Cock Notch **Releasing slide into battery causes hammer to drop to half-cock notch or follow slide forward**	• Hammer hooks stoned shallower than safe 0.018" minimum. • Weak sear leaf on 3-finger flat leaf spring (left sear leaf bent forward too far). • Excessive trigger bounce on heavy steel trigger without trigger stop. • Rounding of the sear nose from improper freehand stoning.	**Gunsmith Action: DANGEROUS CONDITION. Bench test: Pull hammer to full cock, release slide from slide lock on empty chamber. If hammer drops to half cock, replace sear and hammer. Bend left leaf of 3-finger sear spring to increase forward sear pressure. Check hammer hook depth on optical comparator.**

Stoppage 3: Grip Safety Fails to Block Trigger Bow **Trigger can be pulled and hammer dropped without depressing the grip safety**	• Grip safety blocking arm over-filed during custom beavertail installation. • Right leaf of 3-finger spring fatigued (insufficient rearward spring pressure). • Trigger bow overtravel stop screw positioned too far forward.	**Gunsmith Action: DETAIL SAFETY TEST. Cock hammer. With grip safety released, pull trigger firmly; hammer MUST NOT drop. Grip safety must disengage only after at least 1/3 to 1/2 of inward depression. If over-cut, weld-up arm with TIG and hand-file to fit, or replace with new beavertail.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Anvil 075: 1911 Conserve and Reassemble Masterclass	Mark Novak / Anvil Gunsmithing	Master gunsmith Mark Novak performs a complete bench conservation, detail disassembly, flat leaf spring tuning, and reassembly of a classic military 1911A1.
1911 Extractor Tuning, Tensioning & Fitting	Brownells Gunsmithing Bench	Step-by-step method to gauge extractor deflection, beveling the bottom entry corner for feeding, and testing tension with a trigger pull gauge.